Image Inpainting and Object Removal

Week 4 Progress Report: Diffusion Benchmarking (FLUX.1-Fill & PixArt-α), Cross-Paradigm Comparison, and Failure Analysis

Ivan, Aleksandr, Egor, Tatiana | Computer Vision 2026 | October 27, 2026

Abstract

We investigate classical, CNN-based, Transformer-based, and diffusion-based image inpainting approaches for zero-shot object removal on the CORNE-Val benchmark. This Week 4 report summarizes the current project state, completed experiments across all four architectural families, a comprehensive quantitative comparison over six mask configurations, and key diagnostic observations. During this week, we evaluated diffusion-based models and found that the heavyweight `FLUX.1-Fill` architecture suffers from prohibitive inference latency and semantic replacement errors—substituting removed objects with newly hallucinated ones (e.g., turning a fire truck into a red sports car). To complete the full 1,308-run benchmark, we transitioned to a lighter diffusion transformer, `PixArt-α` (`PixArt-XL-2-1024-MS`). Our results demonstrate that while pretrained `LaMa` achieves superior background reconstruction (19.75 dB PSNR on effect-aware masks), unconditioned or zero-shot diffusion models struggle without task-specific object-removal pretraining.

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1 Current Project State & Completed Experiments

The project is structured around two synchronized components: a reproducible **Evaluation Mode** console benchmark built on the paired `CORNE-Val` dataset (218 samples with object-present `shot`, clean ground-truth `bg`, tight `mask_sam`, and shadow-inclusive `mask_eff`), and an interactive **Application Mode** web interface displaying progressive multi-model results in a carousel. At the conclusion of Week 4, the core evaluation pipeline and all four paradigm wrappers under the shared `inpaint(image, mask)` interface are fully implemented and benchmarked.

1.1 Summary of Completed Benchmark Runs

Across Weeks 2–4, we executed 1,308 evaluation runs per model (218 image pairs × 6 mask geometry conditions: `sam_native`, `sam_dilated10`, `sam_dilated20`, `sam_bbox`, `sam_ellipse`, and `eff_native`). Table 1 summarizes the four model paradigms completed to date.

Approach	Architecture / Paradigm	Checkpoint / Setup	Avg. Latency	Benchmark Status
OpenCV Telea	Classical PDE / Fast Marching	`cv2.INPAINT_TELEA` (CPU)	0.014 – 0.025 s	Completed (Week 2, 1,308 runs)
Pretrained LaMa	CNN with Fast Fourier Convolutions	`big-lama` TorchScript (GPU)	0.626 – 0.645 s	Completed (Week 3, 1,308 runs)
ViTInpainter	Vision Transformer (Self-Attention)	Pretrained ViT checkpoint (GPU)	0.275 – 0.315 s	Completed (Week 3, 1,308 runs)
FLUX.1-Fill / PixArt-α	Diffusion Transformer (DiT)	`PixArt-XL-2-1024-MS` (GPU)	3.597 – 3.635 s	Completed (Week 4, 1,308 runs)

Table 1: Summary of all four inpainting paradigms integrated and evaluated on CORNE-Val.

1.2 Transition from FLUX.1-Fill to PixArt-α

In Week 3, our initial attempts to benchmark `FLUX.1-Fill` (a 12-billion-parameter rectified-flow transformer) encountered severe VRAM and compute time bottlenecks. During Week 4, we conducted targeted qualitative and latency tests on `FLUX.1-Fill` using memory offloading and `bfloat16` precision. These tests revealed two critical issues that made full-dataset evaluation impractical and ineffective for pure background reconstruction:

- **Prohibitive benchmarking runtime:** Even with quantization, multi-step denoising at native resolution required several tens of seconds per sample, making 1,308 sequential evaluations computationally infeasible within our hardware budget.
- **Semantic object replacement instead of removal:** Because `FLUX.1-Fill` is heavily trained for text-guided generative object insertion, running it without an explicit background description causes the model to infer plausible foreground objects from the mask silhouette and road context. As illustrated in Figure 1, instead of removing the masked fire truck and filling the gravel intersection, `FLUX.1-Fill` replaces the fire truck with a red sports car.

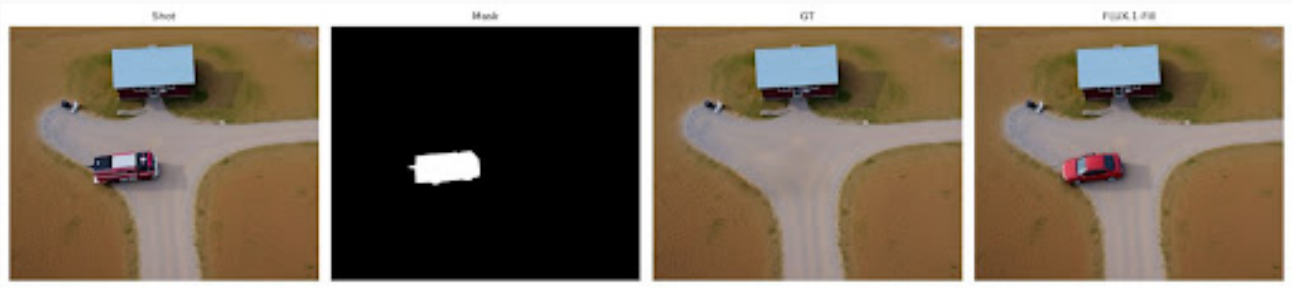


Figure 1: Semantic hallucination in heavyweight diffusion (FLUX.1-Fill). Given the aerial shot and tight mask of a fire truck at a gravel road intersection, FLUX.1-Fill fails to reconstruct the empty road (GT) and instead generates a red sedan/sports car matching the mask footprint and vehicle context.

To complete the full quantitative diffusion benchmark across all six mask conditions on `CORNE-Val`, we replaced `FLUX.1-Fill` with a substantially lighter Diffusion Transformer architecture: **PixArt-α** (`PixArt-XL-2-1024-MS`, ~0.6B parameters) integrated via the Hugging Face Diffusers inpainting pipeline. `PixArt-α` reduced average inference time to ~3.61 seconds per image, enabling full quantitative evaluation across all 1,308 sample-mask pairs.

2 Preliminary Comparison of Approaches

2.1 PixArt-α Quantitative Benchmark

Table 2 presents the complete quantitative metrics for `PixArt-α` across all six mask configurations on `CORNE-Val` (218 samples), reporting both masked-region metrics (primary evaluation criteria) and full-image metrics (secondary check).

Mask Condition	$PSNR_{reg}$ ↑	$SSIM_{reg}$ ↑	$LPIPS_{reg}$ ↓	$PSNR_{full}$ ↑	$SSIM_{full}$ ↑	$LPIPS_{full}$ ↓	Time, s ↓
`sam_native`	14.54 ± 3.98	0.616 ± 0.163	0.453 ± 0.088	26.07 ± 5.14	0.928 ± 0.041	0.097 ± 0.041	3.614 ± 0.145
`sam_dilated10`	14.58 ± 4.02	0.630 ± 0.171	0.464 ± 0.085	25.73 ± 5.15	0.925 ± 0.043	0.105 ± 0.044	3.612 ± 0.045
`sam_dilated20`	14.55 ± 4.08	0.625 ± 0.178	0.476 ± 0.085	25.33 ± 5.15	0.920 ± 0.046	0.114 ± 0.047	3.621 ± 0.036
`sam_bbox`	14.51 ± 4.02	0.543 ± 0.215	0.556 ± 0.092	25.25 ± 5.05	0.918 ± 0.056	0.117 ± 0.060	3.597 ± 0.036
`sam_ellipse`	12.71 ± 3.55	0.517 ± 0.181	0.562 ± 0.078	23.91 ± 4.65	0.916 ± 0.052	0.119 ± 0.055	3.621 ± 0.048
`eff_native`	15.49 ± 4.19	0.654 ± 0.184	0.491 ± 0.086	25.92 ± 4.78	0.919 ± 0.050	0.123 ± 0.053	3.635 ± 0.056

Table 2: PixArt-α (PixArt-XL-2-1024-MS) region and full-image metrics (mean ± std over CORNE-Val, 218 samples).

2.2 Cross-Paradigm Head-to-Head Evaluation

Table 3 consolidates the primary masked-region metrics (`PSNR_reg`, `SSIM_reg`, `LPIPS_reg`) and inference time across all four tested approaches (`OpenCV Telea`, `ViTInpainter`, `Pretrained LaMa`, and `PixArt-α`). This allows a direct comparison across classical, CNN, Vision Transformer, and Diffusion paradigms under identical mask constraints.

Mask Condition	Model	$PSNR_{reg}$ ↑ (dB)	$SSIM_{reg}$ ↑	$LPIPS_{reg}$ ↓	Time, s ↓
`sam_native` (tight object core)	OpenCV Telea (Baseline)	12.74 ± 4.01	0.603 ± 0.146	0.487 ± 0.111	**0.014 ± 0.009**
	ViTInpainter	11.76 ± 3.89	0.503 ± 0.160	0.489 ± 0.123	0.285 ± 0.172
	PixArt-α (Diffusion)	14.54 ± 3.98	0.616 ± 0.163	0.453 ± 0.088	3.614 ± 0.145
	Pretrained LaMa	**15.94 ± 4.96**	**0.657 ± 0.172**	**0.359 ± 0.100**	0.634 ± 0.201
`sam_dilated10` (dilation r=10)	OpenCV Telea (Baseline)	14.65 ± 4.48	0.664 ± 0.147	0.446 ± 0.099	**0.018 ± 0.012**
	ViTInpainter	14.66 ± 4.44	0.641 ± 0.154	0.383 ± 0.099	0.315 ± 0.179
	PixArt-α (Diffusion)	14.58 ± 4.02	0.630 ± 0.171	0.464 ± 0.085	3.612 ± 0.045
	Pretrained LaMa	**16.83 ± 5.34**	**0.670 ± 0.177**	**0.357 ± 0.096**	0.641 ± 0.251
`sam_dilated20` (dilation r=20)	OpenCV Telea (Baseline)	15.26 ± 4.75	0.671 ± 0.149	0.453 ± 0.095	**0.021 ± 0.013**
	ViTInpainter	15.67 ± 5.17	0.649 ± 0.160	0.381 ± 0.093	0.301 ± 0.172
	PixArt-α (Diffusion)	14.55 ± 4.08	0.625 ± 0.178	0.476 ± 0.085	3.621 ± 0.036
	Pretrained LaMa	**17.53 ± 5.79**	**0.672 ± 0.181**	**0.358 ± 0.092**	0.626 ± 0.157
`sam_bbox` (bounding box)	OpenCV Telea (Baseline)	14.45 ± 4.09	0.545 ± 0.196	0.608 ± 0.091	**0.025 ± 0.021**
	ViTInpainter	14.20 ± 4.28	0.466 ± 0.202	0.534 ± 0.090	0.299 ± 0.181
	PixArt-α (Diffusion)	14.51 ± 4.02	0.543 ± 0.215	0.556 ± 0.092	3.597 ± 0.036
	Pretrained LaMa	**17.26 ± 5.13**	**0.588 ± 0.226**	**0.456 ± 0.110**	0.636 ± 0.156
`eff_native` (object + shadow)	OpenCV Telea (Baseline)	17.42 ± 5.40	**0.722 ± 0.146**	0.420 ± 0.099	**0.023 ± 0.016**
	ViTInpainter	18.96 ± 5.61	0.713 ± 0.162	0.332 ± 0.088	0.275 ± 0.169
	PixArt-α (Diffusion)	15.49 ± 4.19	0.654 ± 0.184	0.491 ± 0.086	3.635 ± 0.056
	Pretrained LaMa	**19.75 ± 6.06**	0.711 ± 0.180	**0.328 ± 0.090**	0.642 ± 0.170

Table 3: Cross-paradigm comparison of masked-region metrics on CORNE-Val (note: on `sam_ellipse`, LaMa also leads with 14.94 dB PSNR and 0.483 LPIPS vs. 12.71 dB / 0.562 for PixArt-α, 12.49 dB / 0.605 for Telea, and 11.34 dB / 0.581 for ViTInpainter).

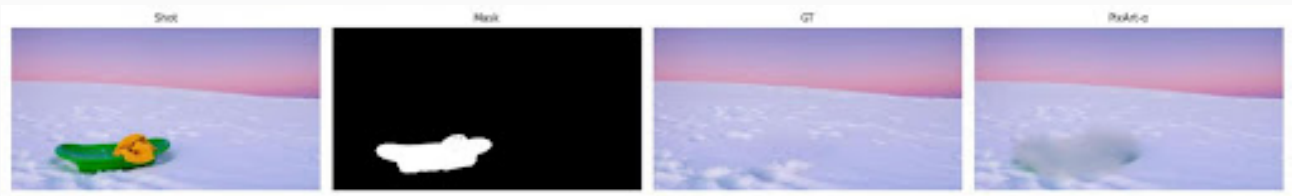


Figure 2: Qualitative output of PixArt-α without task-specific inpainting pretraining. While PixArt-α avoids hallucinating a completely new vehicle like FLUX.1-Fill, it replaces the green sled on snow with an over-smoothed, hazy grey cloud that fails to match the surrounding high-frequency snow footprints (15.49 dB PSNR on `eff_native` vs. 19.75 dB for LaMa).

3 Observations & Failure Analysis

3.1 Four Primary Failure Modes & Experimental Findings

Synthesizing our quantitative tables and visual inspections across Weeks 2–4, we highlight four major technical observations and failure modes that govern object removal performance:

- **Failure Mode 0 — Classical Telea fails on complex objects and structured backgrounds:** Because `OpenCV Telea` relies purely on local PDE boundary marching without semantic priors, it cannot synthesize missing geometric structures or complex textures. When removing large or complex foreground objects, Telea inward-propagates contrasting edge pixels, producing severe directional streaking, blurry color puddles, and broken background geometry (LPIPS degrades up to 0.608 on `sam_bbox`).
- **Failure Mode 1 — Strong sensitivity to mask coverage and unmasked shadow remnants:** Across all models, inpainting quality degrades drastically when the mask fails to cover cast shadows, reflections, or tight boundary pixels (`sam_native`), or conversely when coarse geometric masks (`sam_bbox`, `sam_ellipse`) erase excessive valid context. On tight masks, residual foreground/shadow pixels along the perimeter trick `ViTInpainter` (11.76 dB PSNR) and `LaMa` (15.94 dB) into hallucinating dark blobs or object silhouettes. Switching to effect-aware masks (`eff_native`) boosts PSNR by +7.20 dB for ViTInpainter and +3.81 dB for LaMa.
- **Failure Mode 2 — Heavyweight diffusion (FLUX.1-Fill) exhibits high latency and semantic object substitution:** Large text-to-image rectified-flow models require excessive benchmarking time and strongly favor generating salient foreground objects over empty backgrounds. When conditioned on an object-shaped mask (such as the fire truck in Figure 1) without task-specific removal guidance, `FLUX.1-Fill` acts as an object editor rather than an object remover, hallucinating a red sports car inside the mask.
- **Failure Mode 3 — Lightweight diffusion models (PixArt-α) underperform without dedicated inpainting pretraining:** Repurposing a general text-to-image diffusion backbone (`PixArt-XL-2-1024-MS`) for zero-shot mask inpainting without dedicated object-removal fine-tuning yields flat, washed-out latent reconstructions (Figure 2). Unlike `LaMa` and `ViTInpainter`, which are explicitly trained with high-receptive-field perceptual and adversarial losses for hole filling, `PixArt-α` shows minimal gain on clean effect-aware masks (15.49 dB PSNR, trailing even classical Telea at 17.42 dB) while running ~5.6× slower than LaMa.

Key Experimental Takeaway

Task-specific architectural inductive bias outweighs raw parameter count for zero-shot object removal. **Pretrained LaMa** (Fast Fourier Convolutions) consistently ranks first across every mask configuration in both pixel fidelity (19.75 dB PSNR) and perceptual realism (0.328 LPIPS), while **ViTInpainter** provides the best speed-to-quality trade-off on accurate masks (0.275 s per image, 18.96 dB PSNR).

4 Remaining Work & Next Steps

With the quantitative benchmarking of all four model families completed on `CORNE-Val`, our remaining work focuses on finalizing the interactive web application, visual aggregation, and final presentation deliverables:

Remaining Tasks Before Final Submission

- **End-to-end Application Mode integration:** Connect the shared Python inference backend (serving Telea, LaMa, ViTInpainter, and PixArt-α) to the interactive web frontend so user-uploaded images and manually drawn masks populate the progressive 4-model comparison carousel in real time.
- **Diffusion prompt & dilation refinement (ablation):** Run a small targeted ablation testing whether negative prompting (e.g., suppressing foreground object generation) or automatic mask dilation in Application Mode mitigates boundary artifacts and object hallucinations on user-drawn masks.
- **Final visual plots and documentation:** Generate the final aggregated bar charts (PSNR, SSIM, LPIPS, and log-scale latency across all 6 mask types) from `summary.csv`, finalize the repository documentation at https://github.com/Kazualov/bye-bye-AI, and prepare the final 2-page technical report and live demo presentation.